

one ECU may be responsible for many associated systems in some cases.

On a primary level, an ECU is fed with several inputs, and those inputs are then assessed by the ECU and compared with stored onboard data. The ECU decides what needs to be done to ensure the system in question functions properly and issues new commands to suit. The output alters the operation of the whole system, providing the desired effect.

High-performance computer

High-performance computer (HPC) is another added invention to advanced automotive technology. It not only acts as an application server for traditional vehicle functionalities but is also an essential element of the vehicle as a part of the Internet of Everything. It opens up the possibilities to integrate applications and services over-the-air. Just like a smartphone, the vehicle's functions become updatable and upgradable throughout its lifetime.

With the evolution of electronic systems in vehicles, manufacturers will have a platform to introduce more functionalities to the vehicles that are used as autonomous vehicles and enhance the existing functionalities of the vehicles. An HPC plays an essential role in autonomous vehicle operation by bringing in more opportunities for the automotive industry.

Mechatronics

Mechatronics includes a combination of robotics, electronics, computer, telecommunications, systems, control, and product engineering. It opens up the possibilities which we could not have imagined with traditional mechanical systems.

Taking a simple example of mechatronics, one of the first few subsystems to be automated was the formerly purely mechanical carburetor, which now uses a computer-controlled fuel injector. During a cold-engine start-up, automated fuel injectors adjust fuel-air mixtures in real-time. Mechatronics did not just add motorised controllers to a carburetor, such as the automated tape drive, but used the fuel injectors to handle the old functions better.

Components per car have increased compared to when cars were just a mechanical machine, which has become possible due to mechatronics.

But using computer control over the previously mechanical system means fewer and less-expensive components to outperform tight-spec mechanically calibrated components.

Understanding the changing E/E architecture

The E/E architecture is a convergence of many domains: electronics hardware, network communications, and software applications combined to form the vehicle architecture. And this architecture is moving from distributed ECUs to centralised HPCs. With the given trend, the industry aims to achieve a server-oriented HPC architecture by 2025 and standardised HPC architecture by 2030. But the question arises, why is the industry moving towards this, why is it so critical?

Why change to new architecture: Present challenges

The future demands more flexibility. The future of mobility is connected and autonomous, and it requires a central node that allows necessary data management capabilities for cloud/IoT services. The transformation enables overall E/E-architecture optimisation as well as features upgradability and updatable over the vehicle's lifetime.

Second, as vehicles have become increasingly sophisticated, the demand for raw computing power is also increasing exponentially. The new driving functions entirely require unprecedented amounts of processing power, underwritten by tens of millions of lines of computer code.

Rather than adding to the already excessive quantity of separate ECUs, it has become necessary to completely re-think and re-design the car's central nervous system, consolidating the computing power and allowing the vehicle to operate in a smarter, more efficient manner.

Systems powered by the new E/E architecture

The car's body and cockpit HPC enables newer mobility concepts and improves the consumer-end experience. The car's HPC system makes it possible to update and upgrade the vehicle over its lifetime, just like we keep updating and upgrading our smartphones. The systems in the car

will never get outdated.

Besides, it provides state-of-the-art security features. It also creates a platform for third-party software and service integration. For instance, keyless car access makes your mobile phone turn into your car's key, where you can lock and unlock your car and take care of your car's safety. Such advanced features can only be enabled with the help of HPC.

Another example of HPC use is cross-domain hub. It is a crucial step towards transforming modern vehicle architecture that demands the consolidation of ECUs. Cross-Domain Hub is a high-performance computer that integrates all the displays into a single unit and provides a holistic human-machine interface. The driver can dynamically distribute content across multiple displays and place the information they need to see.

The displays are enabled with touch, gesture recognition, voice recognition, and haptic feedback, allowing the driver to stay focused on the road and make driving safe. It provides a flexible user interface personalised for the user. The user interface is seamless across the integrated display, providing a consistent human-machine interface philosophy.

Way-forward

Increasing functionality and connectivity of vehicles bring traditional distributed architectures with up to hundred or more ECUs. This limits the complexity and ability to manage innovative functions.

Within a new and more centralised architecture design, the HPC replaces conventional distributed ECUs and acts as a central 'electronic brain' to manage data within the vehicle and beyond. The vehicle becomes part of the Internet of Things, and the complexity is simplified by bringing classic vehicle functions into one control unit.

This not only resolves the challenge of getting more power in less space but also gives vehicles the freedom to stay up to date at all times. Thus, the new vehicle architecture is crucial for making the future connected and autonomous vehicles a reality. **EFY**



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